

Total No. of Questions : 8]

SEAT No. :

PC2486

[Total No. of Pages : 3

[6354]-615

B.E. (Mechanical Engineering)

AUTOMOBILE DESIGN

(2019 Pattern) (Semester-VII) (Elective-III) (402044 A)

Time : 2½ Hours]

[Max. Marks :70

Instructions to the candidates:

- 1) Solve Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Use of electronic pocket calculator is allowed.
- 5) Assume suitable data if necessary.

**Q1)** a) A propeller shaft is required to transmit 50 kW power at 600 rpm. It is a hollow shaft, having an inside diameter 0.8 times of the outside diameter. It is made of steel ( $S_{yt} = 380 \text{ N/mm}^2$ ) and the factor of safety is 4. Calculate the inside and outside diameters of the shaft. Assume ( $S_{sy}=0.5S_{yt}$ ) [9]

b) Describe with neat sketch the working of power steering unit. [9]

OR

**Q2)** a) Explain with neat sketch full floating axle and state its applications. [8]

b) Pair of bevel gears with  $20^\circ$  pressure angle, consists of a 20 teeth pinion meshing with a 30 teeth gear. The module is 4 mm, while the face width is 20 mm. The material for the pinion and gear is steel 50C4 ( $S_{ut} = 750 \text{ N/mm}^2$ ). The gear teeth are lapped and ground (Class-3) and the surface hardness is 400 BHN. The pinion rotates at 500 rpm and receives 2.5 kW power from the electric motor. The starting torque of the motor is 150% of the rated torque. Determine the factor of safety against bending failure and against pitting failure. [10]

**Q3)** a) Write a short note on: [9]

- i) Classification of pressed steel disc wheel
- ii) Factors affecting tyre life

b) Explain limitations and benefits of radial ply tyre over the cross ply tyre. [8]

OR

P.T.O.

**Q4) a)** Explain with neat sketch construction and working of hydraulic brake system for a vehicle [9]

**b)** A circular road course track has a radius of 500 m and is banked to  $10^\circ$ . If the coefficient of friction between the road and tyre is 0.25. [8]

Compute:

- i) The maximum speed to avoid slipping and
- ii) The optimum speed to avoid wear and tear on the tyres.

**Q5) a)** Explain the working and construction of the shock absorber. State its advantages. [9]

**b)** Design a Helical coil spring with  $P_{\max} = 7.5$  kN. The mean coil diameter (D) Should be 150 mm from space stiffness of the spring.  $k = 75$  N/mm. Spring made of coil hardened and tempered steel wire ( $S_{ut} = 1250$  N/mm<sup>2</sup>). Permissible shear stress=30% of Ultimate tensile strength ( $S_{ut}$ )  $G = 81370$  N/mm<sup>2</sup>. Assume a Wahl's Stress factor,  $K = 1.213$ , [9]

Calculate:

- i) Spring Index,  $C$
- ii) Wire diameter,  $d$
- iii) Number of active coils,  $N$

OR

**Q6) a)** Explain with neat sketch active suspension system with its limitations and advantages. [9]

**b)** A semi-elliptic leaf spring used for automobile suspension consists of three extra full-length leaves and 15 graduated-length leaves, including the master leaf. The centre-to-centre distance between two eyes of the spring is 1 m. The maximum force that can act on the spring is 75 kN. For each leaf, the ratio of width to thickness is 9:1. The modulus of elasticity of the leaf material is  $207\,000$  N/mm<sup>2</sup>. The leaves are pre-stressed in such a way that when the force is maximum, the stresses induced in all leaves are same and equal to  $450$  N/mm<sup>2</sup>. [9]

Determine:

- i) The width and thickness of the leaves;
- ii) The initial nip; and
- iii) The initial pre-load required to close the gap  $C$  between extra fulllength leaves and graduated-length leaves.

**Q7) a)** Explain the following: [10]

- i) Mechanical packaging
- ii) Automotive industry packaging.

b) List and explain ergonomic considerations in design of automobile. [7]

OR

**Q8) a)** Explain with suitable examples use of anthropometry in automobile design. [9]

b) Write a note on applications of biomechanics in vehicle. [8]

